Kemal Sivro

Software Developer | Full-Stack, AI/ML

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PROFESSIONAL SUMMARY

Software developer and Data Science student at GRAFiT BA, where I re-architected a legacy Microsoft Dataverse estate into 8 independently deployable managed solutions and automated its release path with Azure DevOps CI/CD. Full-stack day to day: .NET/C#, FastAPI and PostgreSQL behind the API, React, React Native and Next.js in front of it, plus Dynamics 365 plugin development. Also a Teaching Assistant at ETF Sarajevo covering Python, C++ and Algorithms and Data Structures. Outside work I build and run my own production systems end to end: a Django study platform with 39 courses and about 625 tests, a Go media cloud on my own server, and LLM, computer vision and reinforcement learning projects.

WORK EXPERIENCE

GRAFiT BA d.o.o. 2025 - present

Software Developer, Zenica, BiH

- Led a ground-up re-architecture of a legacy Microsoft Dataverse and Power Platform estate, consolidating three overlapping unmanaged monolithic solutions into 8 independently deployable managed solutions by separating shared schema, platform tables, PCF controls and cloud flows from per-app business logic so every component has exactly one owner.
- Designed and built a custom internal CLI toolchain of three purpose-built tools covering the full solution lifecycle, from export and unpack through contract-based auditing to guarded dependency-ordered import, replacing manual exports with a repeatable process enforced by machine-checked publisher-prefix, dependency and environment-isolation rules.
- Automated the release path as an Azure DevOps CI/CD pipeline running deployment, cleanup and audit scripts on every promotion across development, UAT and production, catching cross-solution ownership conflicts, missing dependencies and configuration drift before they reached a live environment.
- Authored the accompanying architecture and decision records, capturing packaging and ownership rules as maintained reference material instead of undocumented tribal knowledge.
- Developed Dynamics 365 plugins in .NET/C# (IPlugin) implementing server-side event-driven logic with idempotency guarantees, transactional correctness and structured error handling.
- Delivered the backend for a real-time field dispatch platform by designing a FastAPI service layer and PostgreSQL schema covering job assignment, operator tracking and live status management.
- Built the operator-facing mobile client for the same platform using React Native, implementing job queue views, real-time dispatch updates and field confirmation flows.

Faculty of Electrical Engineering, University of Sarajevo 2025 - present

Teaching Assistant (part-time)

- Deliver lab sessions and structured feedback for Python, C++ and Algorithms and Data Structures coursework to engineering undergraduates.

Various Clients (Freelance) 2024 - 2025

Full-Stack Developer (Remote)

- Designed and shipped custom web applications for remote clients using Next.js, Astro, React and WordPress, managing full project lifecycles from requirements through handoff entirely async.

Inzpire d.o.o. 2024

Full-Stack Development Intern (MEAN Stack), Zenica, BiH

- Built REST API endpoints and frontend components for a MEAN stack application (MongoDB, Express, Angular, Node.js) as part of a collaborative engineering team.

PROJECTS

DSAI Vault | Full-stack study platform, built and run solo | 2026 dsai3.pages.dev

- Self-hosted study site for a university Data Science and AI programme: 39 courses of lecture notes, spaced-repetition flashcards, practice problems, past papers and exam plans, with Google sign-in and per-student progress.
- Content as code: Markdown/YAML validated against Pydantic schemas, with errors naming the exact file and line. The in-browser editor is generated from the same schemas, with SHA-256 content-addressed versioning, diff history and one-click restore.
- About 625 automated tests in GitHub Actions CI/CD, zero-downtime deploys to a single Linux server, Content Security Policy, rate limits and role-based access. Students author courses with an AI-assisted public toolkit.

Python 3.13, Django 6, Pydantic, SQLite, Alpine.js, KaTeX, Docker, Caddy, Cloudflare, GitHub Actions

ListForge | LLM pipeline with deterministic validation | 2026 github.com/vrosi21/list-forge

- Product copy generator for multi-brand e-commerce: a language model drafts each listing, and deterministic, tested code decides whether it is approved, routed to a human reviewer, or failed. Nothing publishes automatically.
- Every generated claim is checked against the product's catalogue row (unsupported numbers, materials, origins, brand-banned claims). Transport retries (backoff with full jitter) are kept separate from content retries that re-ask with the exact validation errors.
- Caught 14 of 14 planted violations with 0 of 6 clean cases flagged, behind 281 backend and 139 frontend tests run by CI on every push.

Python 3.13, FastAPI, Pydantic v2, Groq, SQLite, asyncio, Next.js 16, React 19, TypeScript, Docker, GitHub Actions

Prismo Cloud | Self-hosted media cloud | 2026 github.com/vrosi21/prismo-cloud

- Imgur/Giphy alternative for screenshots, recordings, GIFs and stickers on my own VPS behind Cloudflare. Stateless Go API with presigned direct uploads to content-addressed S3 object storage (Garage).
- PostgreSQL job queue (SKIP LOCKED + LISTEN/NOTIFY) feeds ffmpeg workers that produce thumbnails, MP4 and GIF versions. Sharing by private link or per Google account (view/edit) through Google OIDC.
- Windows desktop helper (Win32 + WebView2) with a hotkey picker that pastes any sticker into any chat app, and in-browser background removal with U2-Net on ONNX Runtime Web.

Go, PostgreSQL, S3/Garage, ffmpeg, Docker, Caddy, Cloudflare Pages/Workers, React 19, TypeScript, Vite, Tailwind

Medical Form OCR Pipeline | Computer vision | 2026 github.com/vrosi21/form-ocr-pipeline

- Four-stage pipeline (fiducial affine registration, CLAHE preprocessing, per-field model routing, weighted fuzzy record linkage) that raised field accuracy on scanned handwritten forms from 20.8% to 98.0%, with 100% patient ID accuracy on held-out data.
- Fine-tuned an EasyOCR CRNN recogniser with CTC loss using an augmentation-safe train/val split and early stopping. Found and fixed EasyOCR returning empty strings on single-character crops by bypassing its detection stage.

Python, PyTorch, EasyOCR, OpenCV, Streamlit

More projects with code and write-ups at vrosi21.pages.dev: BMO (self-hosted AI assistant on n8n), Clarity (reinforcement learning task assignment), cartel detection in public procurement (89.0% ROC AUC), TV image denoising (C++), and more.

EDUCATION

BSc Data Science and Artificial Intelligence 2024 - present
Faculty of Electrical Engineering, University of Sarajevo. Teaching Assistant (concurrent).

Richmond Park International Secondary School, Zenica 2015 - 2019
Honors Student. Gold Medalist in IT competitions.

SKILLS

Languages: Python, TypeScript, JavaScript, C#, C++, Go

Backend: .NET, FastAPI, Django, Node.js, Express, REST APIs, Pydantic

Frontend: React, React Native, Next.js, Angular, Astro, Tailwind

AI/ML: PyTorch, scikit-learn, Pandas, NumPy, LangChain, sentence-transformers, EasyOCR, OpenCV, ONNX Runtime, LLMs, RAG, NLP, Reinforcement Learning

Microsoft Stack: Dynamics 365, Dataverse, Power Platform, Power Automate, PCF controls, Power Platform ALM, managed solutions

DevOps: Azure DevOps, GitHub Actions, CI/CD pipelines, Docker, Caddy, Cloudflare, Linux, Git

Data: PostgreSQL, SQLite, MongoDB, Dataverse, S3 object storage

Spoken languages: English (fluent), German (conversational), Bosnian/Croatian/Serbian (native)